

EDGE SENSOR DEFINITION



SV TCL
An SV Probe Company

Report by SV VN team
Date: 6th Jun 2017

Definition

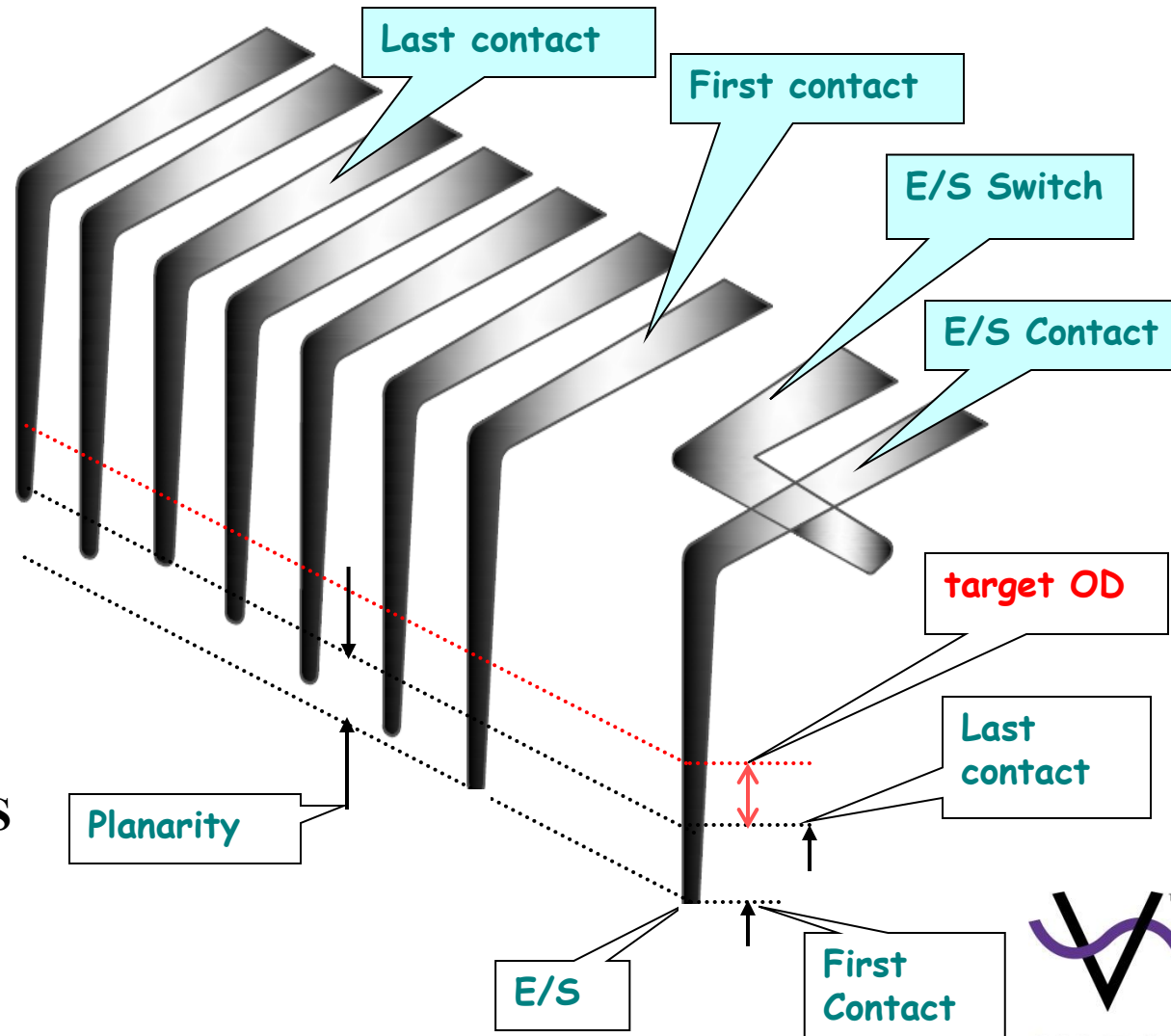
* **OVERDIRVE:** Vertical measurement calculated from the first contact point of the probe to wafer surface.

* **FIRST PROBE OD:** Overdrive of first probe (the lowest probe).

* **LAST PROBE OD:** Overdrive of last probe (the highest probe).

Last probe OD is also the OD Value of Probe Card.

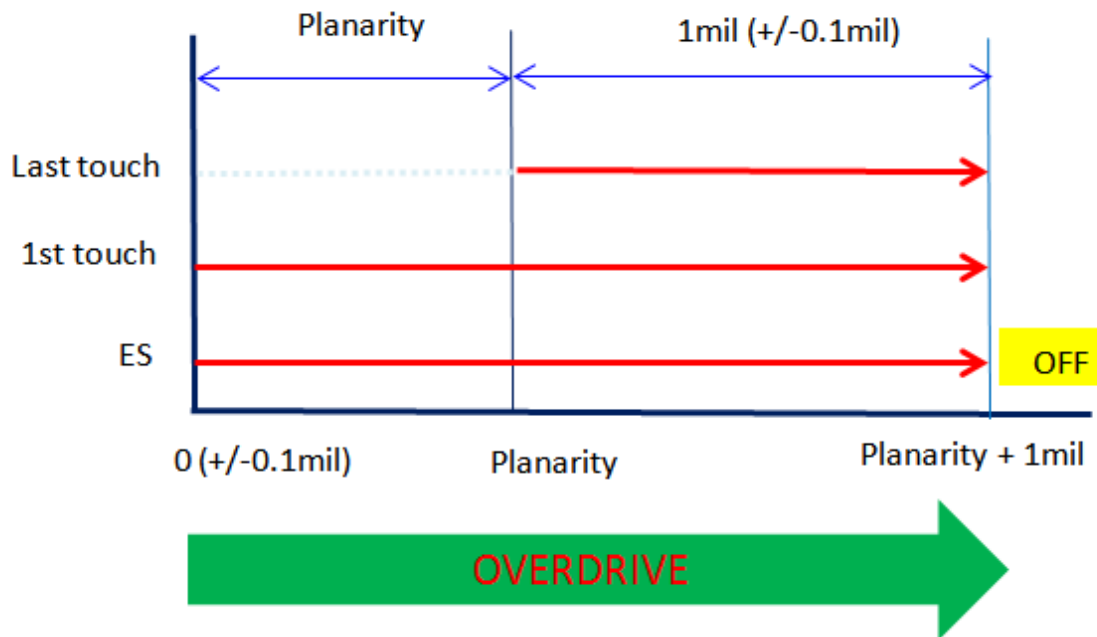
* **E/S OD:** Overdrive of E/S. E/S Switch will open when E/S contacts device surface at this value.



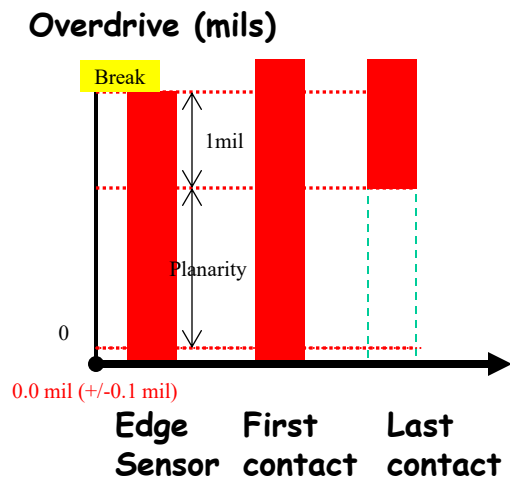
Standard ES

Edge Sensor Contacts at 0.0mil (+/- 0.1 mil) Before First Probe.

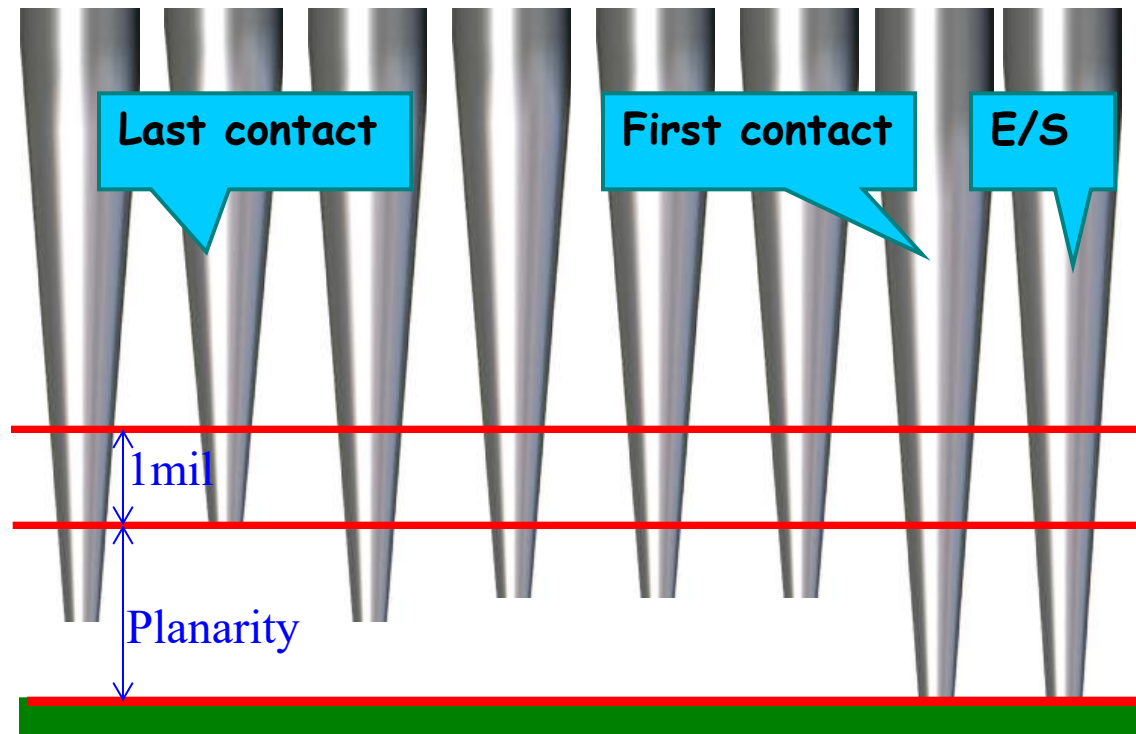
The E/S switch must break or open at 1.0mil (+/- 0.1 mil) after the last probe has contacted the wafer.



Standard ES



Standard ES: Switch Opens at **1.0mil (+/- 0.1 mil)**
After Last contact (SV Standard).



Non- Standard ES

Non-Standard ES:

Switch Opens at **x.xx mils** after Last Probe

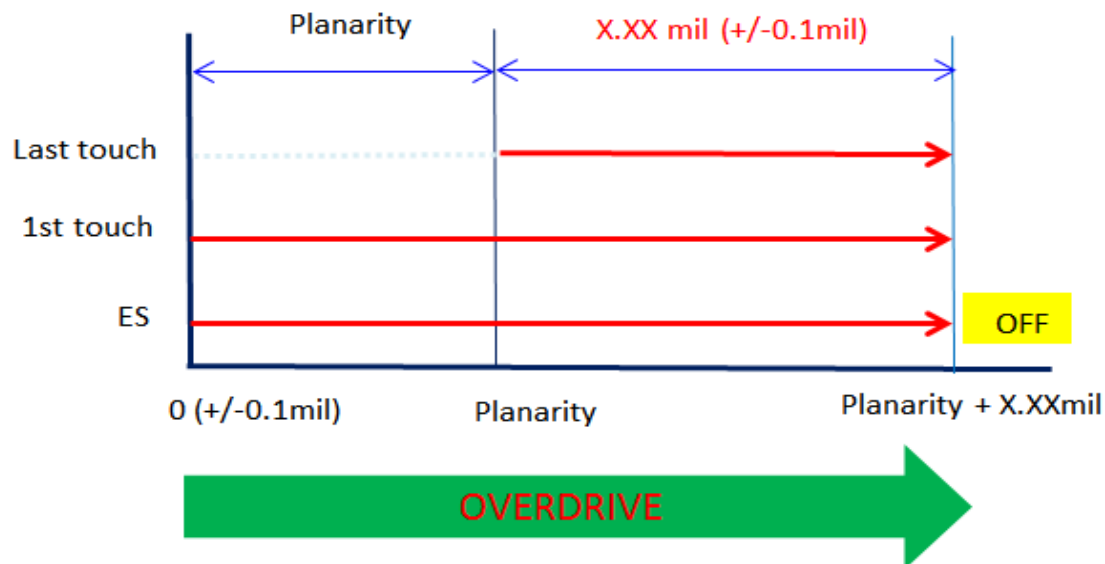
→ Edge Sensor contacts at **0.0mil (+/- 0.1 mil)** before First Probe,

Switch opens at **x.xx (+/- 0.1 mil)** after Last Probe

Ex: Customer requires Switch Opens at 1mils OD.

→ Edge Sensor contacts at **(+/- 0.1 mil)** mils before First Probe

→ Switch opens at **0.9~1.1 mils** after Last Probe.



Non-Standard ES

Overdrive (mils)

Break

$x.xx (+/-0.1 \text{ mil})$

Planarity

0

0.0mil (+/-0.1 mil)

Edge Sensor First contact Last contact

Non-Standard ES: Switch Opens at $x.xx (+/-0.1) \text{ mils}$ After Last contact, it is also OD value of Probe Card

Last contact

First contact

E/S

$x.xx \text{ mil}$

Planarity

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The diagram illustrates the probe tip's position during a measurement. The probe tip is shown as a series of vertical lines. Three blue callout boxes are present: 'Last contact' on the left, 'First contact' in the middle, and 'E/S' (End of Stroke) on the right. Two horizontal red lines are drawn across the probe tip. A vertical blue double-headed arrow indicates the distance between these two lines, labeled 'x.xx mil'. Below this, another vertical blue double-headed arrow is labeled 'Planarity'. The probe tip is shown in a series of positions, with the 'Last contact' position being the leftmost and the 'First contact' position being the rightmost. The 'E/S' position is the rightmost position shown.



Summary

1

The Standard **E/S Contact** Tip must contact the wafer at **0.0mil (+/- 0.1 mil)** before the first contact. The **E/S switch** must break or open at **1.0mil (+/- 0.1 mil)** mils after The last probe has contacted the wafer

2

When a customer specifies a non-standard edge sensor/switch setting, the edge sensor and switch must be on and off on the given specification **x.xx(+/-0.1)mils** after Last contact

3

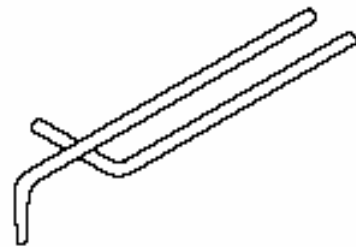
If Probe card has more than 1 E/S, P&A refer to 1st contact to set for the rest E/S

4

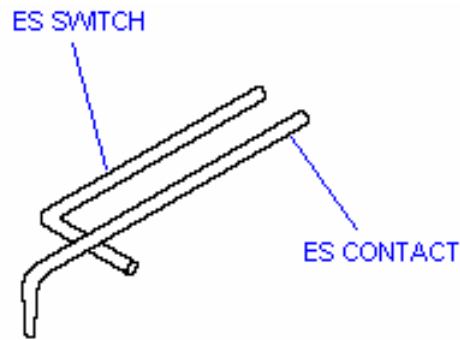
Tolerance: **+/-0.1mil**

Some type of ES

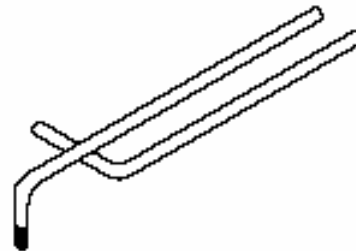
2- WIRES INSOLATED AND 2 WIRE NON-INSULATE (LEFT & RIGHT)



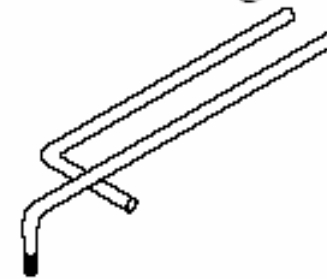
2-wire Left



2-wire Right



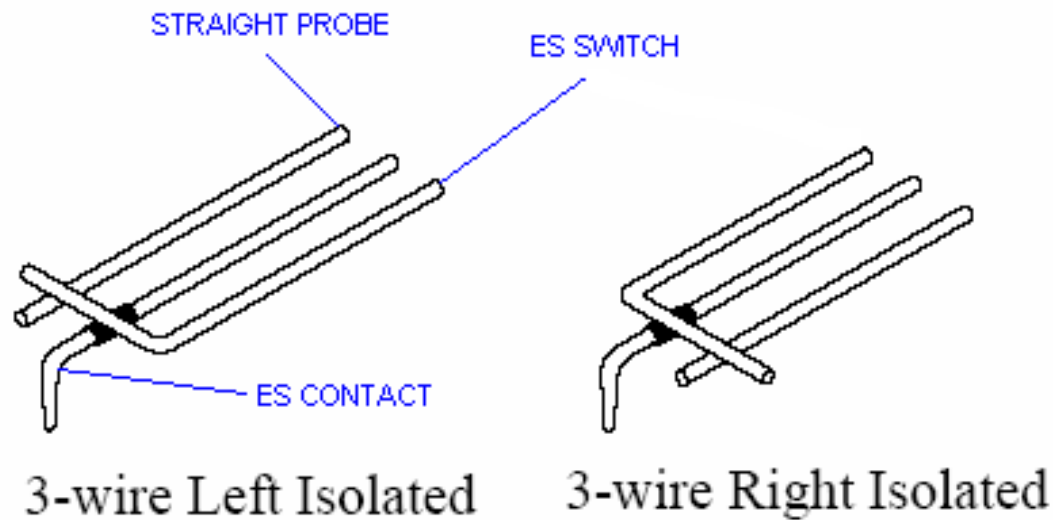
2-wire Left Isolated



2-wire Right Isolated

Some type of ES

3- WIRES INSOLATED (LEFT & RIGHT)





Example#

Eg. 3-Wires ES

